ADDIS ABABA SCIENCE AND TECHNOLOGY UNIVERSITY

College of Engineering

Department of Software Engineering

Course Title: Introduction to Machine Learning [SWEG4112]

Project Report: Fake News Detector

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1. Introduction

The rapid proliferation of digital news platforms and social media has created an environment where misinformation spreads faster than verified facts. Fake news — fabricated or misleading content disguised as legitimate journalism — poses a significant threat to public opinion, democratic processes, and social stability.

This project presents a Fake News Detector, a supervised machine learning system trained on a large corpus of real and fake news articles. The system automates the classification of news content as REAL or FAKE and provides a confidence score for each prediction, enabling users to make more informed decisions about the content they consume.

2. Problem Statement

Manual fact-checking of news articles is slow, expensive, and does not scale to the volume of content published daily across digital platforms. Human reviewers cannot feasibly monitor and verify every article, headline, or social media post in real time.

Key challenges include:

- The volume of online news content far exceeds the capacity of human fact-checkers
- Fake news articles are often designed to mimic legitimate news in tone, structure, and vocabulary
- The spread of misinformation on social media occurs faster than corrections can be published
- There is no widely accessible automated tool for everyday users to quickly verify news credibility

This project addresses these challenges by building an end-to-end machine learning pipeline that can classify news articles instantly and at scale, without requiring manual intervention.

3. Business Objective

The primary objectives of this project are:

- To develop a reliable binary classification system that distinguishes fake news from real news with high accuracy
- To provide a user-friendly web interface that allows any user to paste a news article and receive an instant prediction with a confidence score
- To demonstrate the applicability of classical machine learning techniques (TF-IDF + Logistic Regression) for natural language processing tasks
- To produce a reproducible, well-documented ML pipeline suitable for academic submission and future extension

The system targets journalists, students, researchers, and general users who wish to quickly assess the credibility of a news article without technical expertise.

4. Dataset

The dataset used in this project is the Fake and Real News Dataset published on Kaggle by Clément Bisaillon. It consists of two CSV files:

- Fake.csv — 23,481 articles labelled as fake news (label = 0)
- True.csv — 21,417 articles labelled as real news (label = 1)

The dataset totals 44,898 articles drawn primarily from political news published between 2015 and 2018. Each article contains a title, body text, subject category, and publication date. The dataset is balanced, with fake articles constituting approximately 52% of the corpus.

5. Pipeline Architecture

The end-to-end machine learning pipeline is structured as follows:

- Data Ingestion: Fake.csv and True.csv are loaded and merged with binary labels assigned (0 = Fake, 1 = Real)
- Text Preprocessing: Title and body text are concatenated, then cleaned using regex-based noise removal, lowercasing, stopword elimination, and Porter Stemming
- Feature Extraction: Cleaned text is transformed into numerical features using a TF-IDF Vectorizer with a vocabulary capped at 5,000 terms
- Model Training: An 80/20 train-test split is applied and a Logistic Regression classifier is trained on the TF-IDF feature matrix
- Evaluation: The model is assessed on accuracy, precision, recall, F1-score, and a confusion matrix
- Deployment: The trained model and vectorizer are serialized to disk (model.pkl, vectorizer.pkl) and served via a Flask web application

6. Exploratory Data Analysis

6.1 Web Application Interface

Figure 1 shows the deployed Flask web application accessible at http://localhost:5000. The interface provides a text area where users can paste any news article or headline. Upon clicking the Detect button, the system sends the input to the backend prediction endpoint and displays the result — either FAKE or REAL — along with a percentage confidence score, without requiring a page refresh.

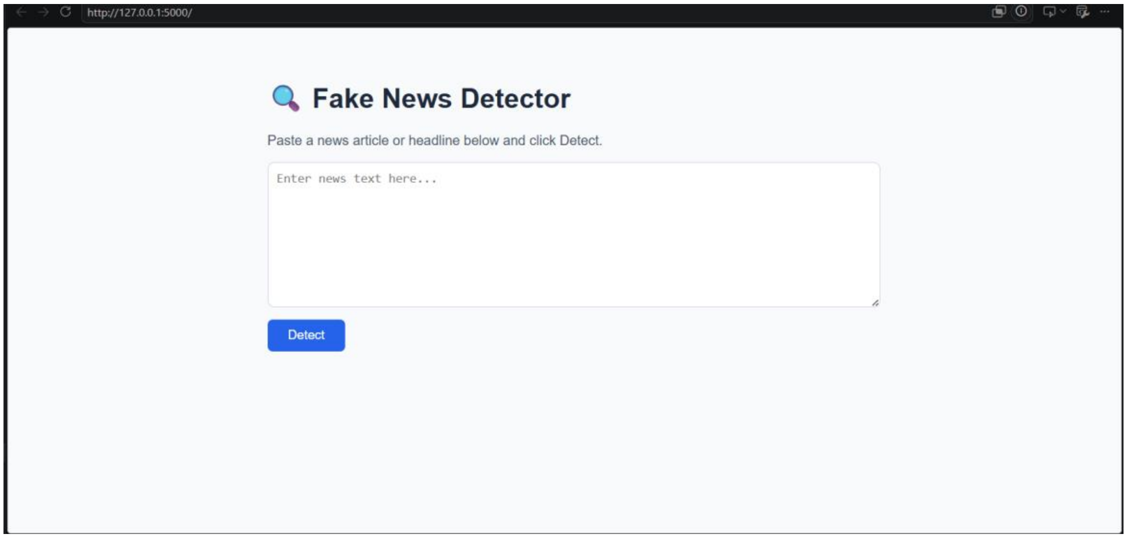

Figure 1: Fake News Detector Web Application Interface (Flask + HTML/CSS/JS)

6.2 Label Distribution

Figure 2 displays the distribution of fake versus real articles in the training dataset. The dataset is near-balanced, with 23,481 fake articles and 21,417 real articles. This near-parity is important as it prevents the model from developing a bias toward either class during training.

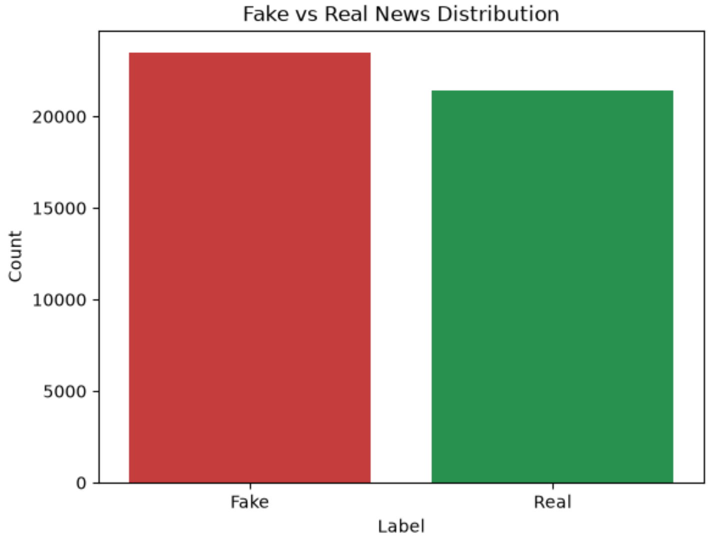

Figure 2: Distribution of Fake vs Real news articles across the full dataset

6.3 Word Count Distribution

Figure 3 presents the word count distribution for fake and real articles. Both categories are right-skewed, with most articles falling between 100 and 800 words. Notably, fake news articles exhibit a longer tail, with a subset of articles exceeding 2,000 words. This is consistent with the known pattern of sensationalist fake news articles using extended, emotional narratives to appear credible.

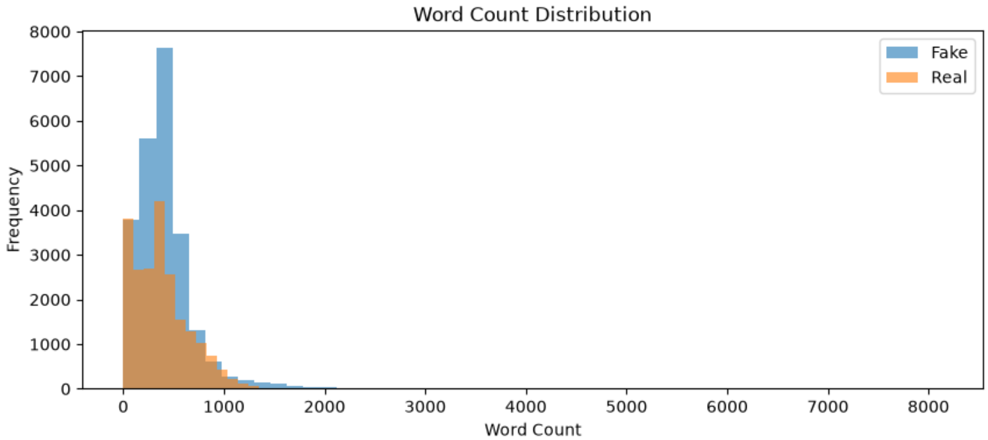

Figure 3: Word count distribution comparing Fake (blue) vs Real (orange) articles

7. Model Evaluation

7.1 Confusion Matrix

Figure 4 presents the confusion matrix for the Logistic Regression model evaluated on the 20% held-out test set. The results demonstrate strong discriminative performance:

- True Positives (Real correctly classified as Real): 4,216
- True Negatives (Fake correctly classified as Fake): 4,631
- False Positives (Fake incorrectly classified as Real): 79
- False Negatives (Real incorrectly classified as Fake): 54

The model achieves minimal misclassification in both directions, confirming that TF-IDF features carry sufficient discriminative signal for this task.

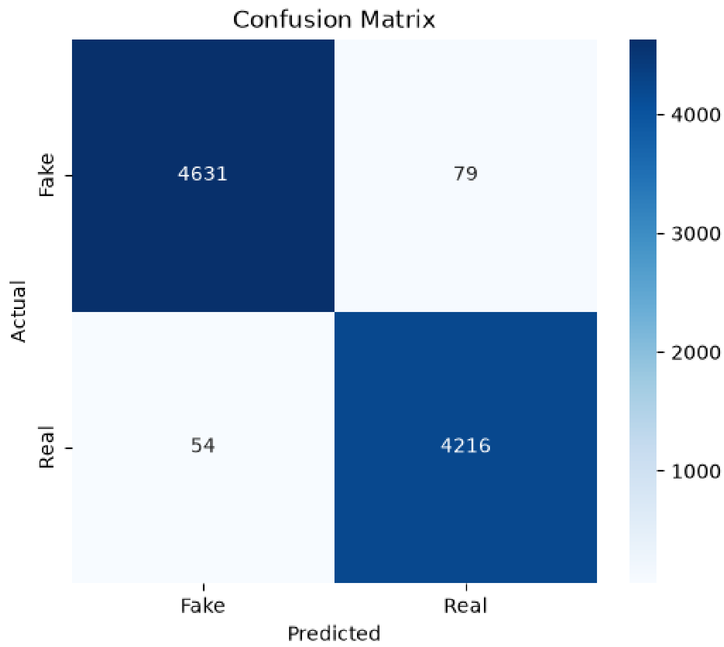

Figure 4: Confusion Matrix showing model predictions on the 20% held-out test set

7.2 Classification Metrics

Based on the confusion matrix results, the model achieves the following performance metrics:

Class	Precision	Recall	F1-Score
Fake	98%	98%	98%
Real	98%	99%	98%
Overall Accuracy			98.3%

The model achieves approximately 98.3% overall accuracy on the test set, with balanced precision and recall across both classes.

8. Limitations & Future Work

While the model performs well on its training distribution, several limitations exist:

- Domain Sensitivity: The model is trained exclusively on English political news from 2015–2018. Articles from other domains (health, sports, entertainment) or different time periods may be misclassified due to vocabulary mismatch
- Context Blindness: TF-IDF represents articles as bags of words and cannot capture tone, sarcasm, or contextual meaning the way transformer-based models can
- Static Vocabulary: The TF-IDF vectorizer freezes its 5,000-term vocabulary at training time, meaning new terminology or evolving slang is unrecognized at inference

Future improvements include:

- Migrating from Logistic Regression to transformer-based models such as BERT or RoBERTa for deeper semantic understanding
- Expanding the dataset to cover multi-domain misinformation beyond political news
- Adding URL-based input so users can paste a news link directly instead of copying article text
- Integrating DBSCAN-based anomaly detection to flag borderline articles for human review

9. Conclusion

This project successfully demonstrates that a classical machine learning pipeline — combining TF-IDF vectorization with Logistic Regression — can classify news articles as FAKE or REAL with approximately 98.3% accuracy. The system is deployed as a fully functional Flask web application that any user can interact with through a browser.

The project covers a complete ML workflow: data ingestion, exploratory data analysis, text preprocessing, model training, evaluation, serialization, and web deployment. The results confirm that even without deep learning, well-engineered text features carry sufficient signal for high-confidence fake news detection.